

You may use a calculator. Read all directions carefully, and write your answers in the space provided. To receive full credit, you must show all of your work.

Part I: Multiple Choice (4 points each)

1. Determine the domain of the function $g(x) = \ln(4 - 3x)$

A. $(\frac{4}{3}, \infty)$
B. $(-\infty, -\frac{4}{3}]$
C. $[\frac{4}{3}, \infty)$
D. $(-\infty, \frac{4}{3})$
E. $(-\infty, \frac{4}{3}]$

$x \neq 0$
 $x > 0$
 $\ln(x)$

$$4 - 3x > 0$$

$$-4 \quad -4$$

$$\frac{-3x > -4}{-3} \rightarrow x < \frac{4}{3} \rightarrow (-\infty, \frac{4}{3})$$

2. For function $f(x) = x^2 - 1$ and $g(x) = \sqrt{3 - x}$, calculate $(f \circ g)(x)$.

A. $2 - x$
B. $\sqrt{4 - x^2}$
C. $x^2 - 4$
D. $x - 2$
E. $\sqrt{x^2 - 4}$

$$(f \circ g)(x) = (\sqrt{3 - x})^2 - 1$$

$$= 3 - x - 1 = -x + 2$$

$$= 2 - x$$

3. Evaluate: $\lim_{x \rightarrow 4} \frac{12x + 2}{7x - 4}$

A. 0
B. 2
C. 4
D. $\frac{25}{12}$
E. Does Not Exist

$$= \frac{12(4) + 2}{7(4) - 4} = \frac{50}{24} = \frac{25}{12}$$

4. Evaluate: $\lim_{x \rightarrow 2} \frac{x^2 + 3x - 10}{x - 2}$

A. 1
B. 3
C. 7
D. Does Not Exist
E. 5

$$= \lim_{x \rightarrow 2} \frac{(x - 2)(x + 5)}{x - 2} = \lim_{x \rightarrow 2} x + 5$$

$$= 2 + 5 = 7$$

5. Evaluate: $\lim_{x \rightarrow -6} \frac{2x + 12}{|x + 6|}$

A. -2
B. -1
C. $\frac{1}{2}$
D. 2
E. Does Not Exist

$$= \lim_{x \rightarrow -6} \frac{2x + 12}{x + 6} = \lim_{x \rightarrow -6} \frac{2(x + 6)}{-(x + 6)}$$

$$= \lim_{x \rightarrow -6} \frac{2}{-1} = -2$$

6. Calculate the value of k which makes $g(x)$ continuous.

$$g(x) = \begin{cases} kx^2 + 3x + 1, & \text{if } x \geq 1 \\ \frac{k}{2} - 5x, & \text{if } x < 1 \end{cases}$$

- A. 6
 B. -18
 C. 18
 D. $-\frac{2}{3}$
 E. 2

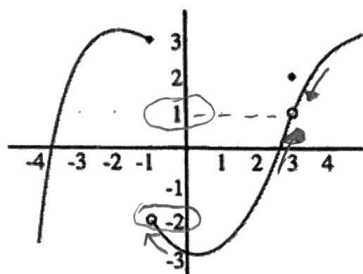
$$kx^2 + 3x + 1 = \frac{k}{2} - 5x \text{ when } x=1$$

$$(1)^2 \quad (1) \quad (1)$$

$$k + 3 + 1 = \frac{k}{2} - 5 \rightarrow \frac{k}{2} = -9$$

$$k + 4 = \frac{k}{2} - 5 \rightarrow k = -18$$

7. Determine the values of A and B (if they exist) using the graph of $f(x)$.



$\lim_{x \rightarrow -1^+} f(x) = A$ and $\lim_{x \rightarrow 3} f(x) = B$

- A. $A = 3, B = 2$
 B. $A = 3, B = 1$
 C. $A = -2, B = 2$
 D. $A = -2, B = 1$
 E. A Does Not Exist, $B = 2$

what is y approaching
as $x \rightarrow ?$

8. Evaluate $\lim_{x \rightarrow \infty} \frac{x^2}{x^4 + 1}$

- A. 2
 B. 5
 C. 1
 D. ∞
 E. 0

longway $\lim_{x \rightarrow \infty} \frac{x^2}{x^4 + 1} \cdot \frac{\frac{1}{x^2}}{\frac{1}{x^2}} = \lim_{x \rightarrow \infty} \frac{1}{x^2 + \frac{1}{x^2}}$

shortway $\lim_{x \rightarrow \infty} \frac{x^2}{x^4}$
 consider highest power terms $= \lim_{x \rightarrow \infty} \frac{1}{x^2} = 0$

lower deg of Top/Bottom

$$= \lim_{x \rightarrow \infty} \frac{1}{x^2 + \frac{1}{x^2}} = \frac{1}{\infty^2 + \frac{1}{\infty^2}} = \frac{1}{\infty^2} = 0$$

9. Evaluate: $\lim_{x \rightarrow \infty} \frac{x^3 + 4x - 2}{6 - 2x^2}$

- A. Does Not Exist
 B. ∞
 C. $-\frac{1}{2}$
 D. $-\infty$
 E. 0

longway $\lim_{x \rightarrow \infty} \frac{x^3 + 4x - 2}{6 - 2x^2} \cdot \frac{\frac{1}{x^2}}{\frac{1}{x^2}} = \lim_{x \rightarrow \infty} \frac{x + \frac{4}{x} - \frac{2}{x^2}}{\frac{6}{x^2} - 2}$

lower deg of Top/Bottom

Graph

$$= \lim_{x \rightarrow \infty} \frac{x + \frac{4}{x} - \frac{2}{x^2}}{\frac{6}{x^2} - 2} = \frac{\infty + \frac{4}{\infty} - \frac{2}{\infty^2}}{\frac{6}{\infty^2} - 2} = \frac{\infty}{-\infty} = -\infty$$

short way

$$\lim_{x \rightarrow \infty} \frac{x^3}{-2x^2} = \lim_{x \rightarrow \infty} \frac{x}{-2} = \frac{\infty}{-2} = -\infty$$

10. Find the horizontal asymptote(s) for the curve $y = \frac{6x}{(16x^4+1)^{\frac{1}{4}}}$

- A. 3
B. -3
C. ± 3
D. 0
E. $\frac{3}{2}$

HA
 $y = \lim_{x \rightarrow \infty} \frac{6x}{(16x^4+1)^{\frac{1}{4}}} = \lim_{x \rightarrow \infty} \frac{6x}{(16x^4)^{\frac{1}{4}}} = \lim_{x \rightarrow \infty} \frac{6x}{\pm 2x}$
even power
 $\sqrt{}, \sqrt[4]{}, \sqrt[5]{}, \dots$
 $= \lim_{x \rightarrow \infty} \pm \frac{6}{2} = \pm 3$

11. Find the equation for the line that is tangent to the graph $f(x) = \frac{1}{x}$ at $x = 2$

- A. $y = -\frac{1}{4}x$
B. $y = \frac{1}{4}x + 1$
C. $y = \frac{1}{4}$
D. $y = -\frac{1}{2}x + \frac{3}{2}$
E. $y = -\frac{1}{4}x + 1$

$m = f'(x) = -1x^{-2}$

$f'(2) = -1(2)^{-2} = -\frac{1}{2^2} = \boxed{-\frac{1}{4}}$

$x = 2$ $y = f(2) = \frac{1}{2}$

$y = mx + b$
 $\frac{1}{2} = (-\frac{1}{4})(2) + b$
 $\frac{1}{2} = -\frac{1}{2} + b$
 $b = \boxed{1}$

12. Which of the following limits represents the derivative of $f(x) = \sin(2x+1)$

- A. $\lim_{h \rightarrow 0} \frac{\sin(2x+h+1) - \sin(2x+1)}{h}$
B. $\lim_{h \rightarrow 0} \frac{\sin(2x+h+2) - \sin(2x+1)}{h}$
C. $\lim_{h \rightarrow 0} \frac{\cos(2x+h+1)}{h}$
D. $\lim_{h \rightarrow 0} \frac{\sin(2x+2h+1) - \sin(2x+1)}{h}$
E. $\lim_{h \rightarrow 0} \frac{\sin(x+h+\frac{1}{2}) - \sin(x+\frac{1}{2})}{h}$

$f(x+h) = \sin(2(x+h)+1)$
 $= \sin(2x+2h+1)$

$\lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$

13. Using the squeeze theorem, determine the limit: $\lim_{x \rightarrow 0} x^2 \sin(\frac{1}{x})$

- A. 0
B. -1
C. 1
D. Does Not Exist
E. $\frac{1}{2}$

$-1 \leq \sin(\frac{1}{x}) \leq 1$
 $\cdot x^2 \quad \cdot x^2 \quad \cdot x^2$

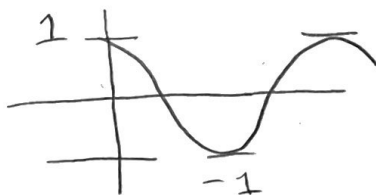
$-x^2 \leq x^2 \sin(\frac{1}{x}) \leq x^2$

$\lim_{x \rightarrow 0} -x^2 = 0$

$\lim_{x \rightarrow 0} x^2 = 0$

14. Determine the limit: $\lim_{x \rightarrow \infty} \cos(x)$

- A. ∞
B. Does Not Exist
C. -1
D. $-\infty$
E. 1



$\Rightarrow \lim_{x \rightarrow 0} x^2 \sin(\frac{1}{x}) = 0$

15. Given that $\lim_{x \rightarrow 2} f(x) = 4$, $\lim_{x \rightarrow 2} g(x) = -2$, and $\lim_{x \rightarrow 2} h(x) = 0$, determine the limit: $\lim_{x \rightarrow 2} \left[\frac{-3h(x) + 5g(x)}{f(x)} \right]^2$

- A. $-\frac{25}{4}$
B. $\frac{5}{2}$
C. $\frac{25}{4}$
D. $\frac{25}{2}$
E. $\frac{5}{4}$

$= \left[\frac{-3(0) + 5(-2)}{4} \right]^2$

$= \left[\frac{-10}{4} \right]^2 = \frac{100}{16} = \frac{25}{4}$

Part II: Free Response (8 points each)

16. Determine the equation of the tangent line to the graph of $y = g(x)$ at $x = -2$ provided $g(-2) = -8$ and $g'(-2) = 12$. (Give your answer in slope intercept form)

$$x = -2 \quad y = g(-2) = -8 \quad m = g'(-2) = 12$$

$$y = mx + b$$

$$-8 = 12(-2) + b$$

$$-8 = -24 + b$$

$$+24 \quad +24$$

$$16 = b$$

$$y = 12x + 16$$

17. a) Write the definition of derivative.

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

- b) Let $f(x) = 2\sqrt{x} + 1$. Using the definition of derivative, calculate $f'(x)$. You must show your work. (No credit will be given for answers based on the fast rule.)

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \quad f(x+h) = 2\sqrt{x+h} + 1$$

$$= \lim_{h \rightarrow 0} \frac{2\sqrt{x+h} + 1 - 2\sqrt{x} - 1}{h}$$

$$= \lim_{h \rightarrow 0} \frac{(2\sqrt{x+h} - 2\sqrt{x})}{h} \cdot \frac{(2\sqrt{x+h} + 2\sqrt{x})}{(2\sqrt{x+h} + 2\sqrt{x})}$$

$$= \lim_{h \rightarrow 0} \frac{4(x+h) - 4x}{h(2\sqrt{x+h} + 2\sqrt{x})} = \lim_{h \rightarrow 0} \frac{4x + 4h - 4x}{h(2\sqrt{x+h} + 2\sqrt{x})}$$

$$= \lim_{h \rightarrow 0} \frac{4h}{h(2\sqrt{x+h} + 2\sqrt{x})} = \lim_{h \rightarrow 0} \frac{4}{(2\sqrt{x+h} + 2\sqrt{x})} = \frac{4}{2\sqrt{x} + 2\sqrt{x}}$$

$$= \frac{4}{4\sqrt{x}} = \boxed{\frac{1}{\sqrt{x}}}$$

18. If a ball is thrown into the air with a velocity of $40 \frac{\text{ft}}{\text{s}}$, its height (in feet) after t seconds is given by $y = 40t - 16t^2$. Find the velocity of the ball when $t = 2$.

$$\text{height} = y = 40t - 16t^2$$

$$\text{velocity} = y' = 40 - 32t$$

$$v(2) = 40 - 32(2) = 40 - 64 = -24 \text{ ft/s}$$

19. Evaluate the limit: $\lim_{x \rightarrow 0} \frac{\sin(3x)}{\sin(5x)} = \boxed{\frac{3}{5}}$

Identity

$$\lim_{x \rightarrow 0} \frac{\sin(ax)}{\sin(bx)} = \frac{a}{b}$$

proof using L'Hopital's rule by derivatives

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{\sin(ax)}{\sin(bx)} &= \lim_{x \rightarrow 0} \frac{\cos(ax) \cdot a}{\cos(bx) \cdot b} \\ &= \frac{\cos(a \cdot 0) a}{\cos(b \cdot 0) b} = \frac{\cos(0) a}{\cos(0) b} = \frac{1a}{1b} = \frac{a}{b} \end{aligned}$$

$\uparrow$ chain rule

(if you haven't had calc yet you may not have heard of this yet Don't worry!)